A frozen whole-brain *Drosophila* connectome in a closed sensorimotor loop

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Abstract

The synapse-resolution *Drosophila* connectome permits simulation of an entire brain with essentially no free parameters, but such models are at present driven open loop: sensory neurons are stimulated and motor neurons are read. We embed a frozen FlyWire FAFB v783 connectome (139,255 neurons; 2,710,038 edges at a five-synapse threshold) in a *closed* sensorimotor loop with a real-time video game, at 1.01× realtime on one consumer GPU. Nothing is trained, and the only fitted quantity is a single global synaptic gain calibrated on a non-visual behaviour. We find a dissociation between modalities. Chemosensory transformations transfer: sugar stimulation drives the proboscis motor pool to 77 Hz and co-stimulation with bitter is 99% suppressed, neither of which was fitted, and an added olfactory channel raises lateral horn activity from 0.32 to 14.1 ± 5.5 Hz and suppresses a descending neuron one synapse downstream with the sign the wiring predicts (-17.7 ± 7.4 Hz), while a degree-preserving shuffle of the same graph does not. Visual motion computation does not transfer. Direction selectivity and looming discrimination both fail: across an operating-point sweep $|\mathrm{DSI}| \leq 0.0121$, some 40× below the value at which experimental studies merely *select* a terminal as direction-tuned, and with a mirror-pair sign relation no better than chance. An earlier version of this manuscript attributed that failure to the model class, arguing that $\mathrm{thr}(A + B)$ is symmetric in A and B and that temporal order is therefore not recoverable. We withdraw the argument. It describes a *memoryless* unit, whereas a two-input correlator built from the same neuron model in the same simulator, differing only in that one arm is delayed, reaches $\mathrm{DSI} = -0.79 \pm 0.017$; conduction delays and synaptic time constants supply exactly the memory the argument omits. The model class is not the obstacle. The correlator geometry is verifiably present in the graph, and the arms are measurably imbalanced — `T4a`'s fast excitatory arm is starved 37 : 1, and the looming detector `LPLC2` sits below its resting potential under roughly tenfold inhibition — yet neither rebalancing the arms with short-term depression nor transplanting per-type gains from an optimized model raises selectivity; both lower it. Candidate mechanisms spanning pooling, sampling geometry, phase offset, saturation, operating point and arm imbalance were tested and eliminated one by one. Two model choices do account for part of the deficit, and both are ours rather than the connectome's: `T4`'s summed synaptic weight sits at the threshold for producing any output, so the cell fires only on tonic bias that carries no stimulus information; and modelling it as a graded unit replaces the expansive nonlinearity a correlator needs with a linear ramp, costing 10.6× against an otherwise identical spiking cell. Correcting both raises the fraction of selective cells from 2.8% to 12.3% and still does not yield direction selectivity: the preferred direction continues to follow the stimulus rather than the wiring. We therefore report the motion failure as mechanistically *unresolved* rather than explained. Read against connectome-constrained models whose unknown parameters are optimized, the wiring is common to both, so what optimization supplies is some combination of per-type gain and timing that we can bound but cannot yet identify — transplanting its gains alone does not suffice. We additionally report a methodological result — a degree-preserving shuffle control is invalid in closed loop, because the two arms generate different stimulus distributions.

Keywords: connectomics; *Drosophila*; whole-brain simulation; closed-loop embodiment; elementary motion detection; olfactory valence; negative results

1 Introduction

An electron-microscopy reconstruction of an adult *Drosophila* brain now provides every chemical synapse between all 139,255 neurons [Dorkenwald et al., 2024], together with cell-type annotation [Schlegel et al., 2024] and per-synapse neurotransmitter prediction [Eckstein et al., 2024]. This raises a question that was previously unaskable: *how much of an animal's behaviour is implied by its wiring alone?*

Two recent lines of work answer parts of it, and they answer differently. Shiu et al. [2024] built a whole-brain leaky integrate-and-fire model with a uniform parameter set and a single fitted scalar — the per-synapse weight W_{syn} — and recovered gustatory behaviours such as proboscis extension. Lappalainen et al. [2024] built a connectome-*constrained* model of 64 optic-lobe cell types in which the unknown biophysical parameters were *optimized* by gradient descent on a motion-estimation task, and recovered single-neuron responses matching 26 experimental studies. These differ in two ways at once: whole-brain versus optic lobe, and frozen versus optimized.

Both are driven open loop. Wang-Chen et al. [2024] close the loop, embedding the optimized visual network of Lappalainen et al. [2024] in a biomechanical fly with reinforcement-learned controllers. The configuration that has not been tested is the remaining corner: a *whole-brain, frozen, unoptimized* connectome placed in a closed sensorimotor loop.

That corner is informative precisely because of the contrast. If a frozen whole-brain model reproduces a computation, the wiring plus a global gain sufficed. If it fails at a computation that the optimized model achieves on the same wiring, the difference isolates what optimization supplied.

We report such a test. We drive the frozen connectome with the rendered frames of *Doom* [Kempka et al., 2016], read eight characterized descending neurons, and return their activity to the game as control input, in real time. The environment is adversarial and non-stationary but — crucially for interpretation — entirely outside the fly's evolutionary experience, so nothing here is a test of ecological competence. It is a test of which sensorimotor transformations survive the transfer from wiring diagram to running system.

Contributions.

1. A real-time closed-loop embodiment of a frozen whole-brain connectome (1.01× realtime, 139,255 neurons, single consumer GPU), released as open code.
2. A modality dissociation: chemosensory transformations transfer, visual motion computations do not.
3. A mechanistic account of the visual failure, with six candidate remedies implemented and measured, isolating a 37 : 1 imbalance in the elementary motion detector's two arms.
4. A positive result that beats a degree-preserving shuffle: an olfactory channel reaches descending neurons through the anatomically named route with the wiring-predicted sign.
5. A methodological caution: shuffle controls are invalid in closed loop.

2 Results

2.1 Model and closed-loop configuration

We load FlyWire FAFB v783 and retain edges of at least five synapses, giving 2,710,038 edges carrying 31,578,726 synapses among 134,013 connected neurons (Table 1). Signs follow the predicted transmitter, with glutamate treated as *inhibitory* — it acts through GluCl chloride channels in flies [Cleland, 1996] and accounts for 16.6% of the edges we simulate (19.5% before thresholding), so importing the vertebrate convention would mis-sign about a sixth of the brain.

We sign per edge rather than per neuron, and in this pipeline the choice is not free. Transmitter is predicted per synapse and reported per neuropil, so one neuron can carry different predictions in different neuropils: 42,078 of the 128,972 presynaptic neurons in the simulated graph (32.6%) carry more than one, and 26,781 (20.8%) carry both signs. Collapsing each neuron to a synapse-weighted majority — the usual way of imposing Dale's law — would flip 3.1% of edges and 2.6% of synapses. This matters for reading the literature as much as for the model: the alternative Princeton release has

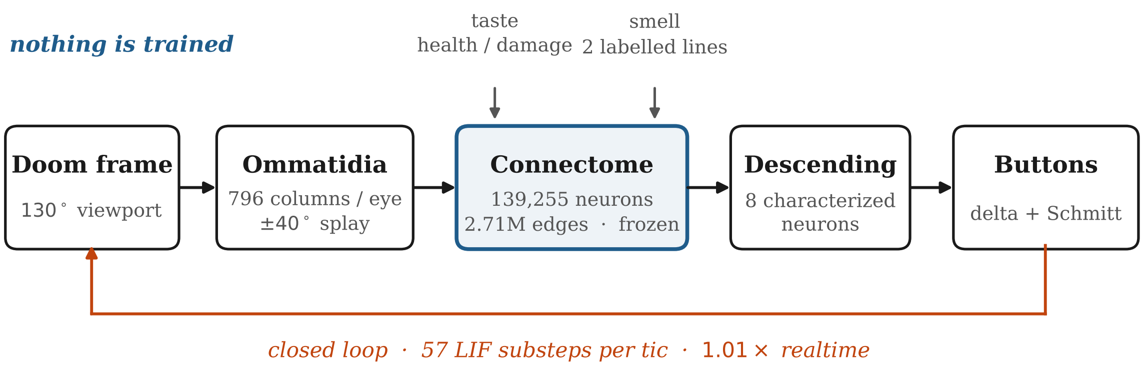

Figure 1. The closed loop. Rendered frames are sampled by the real ommatidial lattice, propagated through the frozen connectome for 57 leaky integrate-and-fire substeps per game tic, and read out at eight characterized descending neurons whose activity becomes the next action. Taste and smell enter as direct drive on gustatory and olfactory receptor neurons. No component is trained and no parameter is fitted to the environment.

Table 1. Graph as simulated. Unthresholded synapse count matches the figure published for this release, confirming the extraction.

Neurons (annotated)	139,255
Edges, unthresholded	16,847,997
Synapses, unthresholded	54,492,922
Edges simulated (≥ 5 synapses)	2,710,038
Synapses simulated	31,578,726
Neurons with ≥ 1 edge	134,013
Graded (non-spiking) units	77,873
Resident GPU memory	31 MB
Wall-clock per 0.5 ms step	$\sim$300 μs
Closed-loop rate	1.01× realtime

Dale's law applied upstream (0 of its 139,003 presynaptic neurons carry more than one transmitter), so per-neuron and per-edge signing coincide there and not here, and transmitter statistics quoted from one release do not describe the other.

Photoreceptors required an explicit override. They release histamine [Hardie, 1989], which is not among the six transmitters the classifier predicts; on `R1-6` output edges the classifier returns ACh 71.2%, Glu 20.2%, GABA 5.8%, and every excitatory call yields the wrong sign for a histaminergic chloride synapse. We force all photoreceptor output inhibitory, flipping 11,920 of 18,437 edges. Without this the optic lobe processes a contrast-inverted image.

Neurons are leaky integrate-and-fire with the parameters of Shiu et al. [2024], themselves drawn from Kakaria and de Bivort [2017]: $V_{\text{rest}} = -52\,\text{mV}$, $V_{\text{thresh}} = -45\,\text{mV}$, $\tau_{\text{mem}} = 20\,\text{ms}$, $\tau_{\text{syn}} = 5\,\text{ms}$, $t_{\text{refrac}} = 2.2\,\text{ms}$, $t_{\text{delay}} = 1.8\,\text{ms}$. We integrate the linear subthreshold system exactly rather than by forward Euler, which we measured 6% low on a single postsynaptic potential. At $dt = 0.5\,\text{ms}$ this gives 57 substeps per game tic, with the frame held across all 57.

Vision enters through the real retinotopy: `column_assignment` supplies 796 ommatidial columns per eye on a hexagonal lattice. Each column samples the rendered frame through a Gaussian acceptance function, implemented as a separable pre-blur followed by bilinear sampling at that column's gaze direction. The engine draws a planar perspective projection, so a column at azimuth θ is sampled at screen coordinate $\tan\theta / \tan(\text{FOV}/2)$ rather than linearly in θ; the linear map misplaces a column at 10° of gaze by 87% and one at 30° by 71%, converging only at the very edge of the viewport. Since retinotopy is the substrate every motion computation runs on, and a warp of that size makes the angular spacing between neighbouring columns vary about twofold across the field, we note it as a correctness requirement rather than a refinement: rigid motion of the world then sweeps the retina at a speed that depends on where one looks, which is precisely the input a correlator tuned to a fixed spacing cannot use. The two eyes are splayed $\pm 40°$; collapsing them onto the screen centre makes both eyes see the same image and drives the left–right steering difference to zero by construction. The engine's field of view must be set through the console rather than the command line — passing `+fov` is silently ignored, and renders at 90°, 130° and 160° are bit-identical — and it resets with each episode. At 130° the viewport covers 64% of the lattice and at 160°, 80%; remaining columns are filled with the frame's mean luminance, because a dark surround is a permanent high-contrast edge that the looming detectors report as a stationary object.

Frames are not held across the substeps of a tic. Doom draws at 35 Hz and the simulator runs 57 substeps per frame, so a held frame presents the retina with a step change followed by 56 substeps of exactly zero temporal derivative, while the delayed arm of the correlator is 80 ms, or three frames — that is, an impulse train aliased against the detector's own delay line. Because a fly in a room receives continuous motion and the strobe is an artifact of the game engine rather than of the biology, we interpolate luminance linearly between consecutive frames across the substeps. This also makes the retina's Weber adaptation advance once per substep as its time constant assumes; stepping it once per tic instead gave it an effective τ of 14 s in place of 0.25 s.

Output is read from eight characterized descending neurons — `DNa02` for steering [Rayshubskiy et al., 2025], `BPN` for forward walking [Bidaye et al., 2020], `MDN` for backward walking [Bidaye et al., 2014], `DNp01` (the giant fibre) for escape [von Reyn et al., 2014] — plus the proboscis motor pool and gustatory receptor neurons. The graded channels use the engine's delta buttons; binary actions use a two-threshold Schmitt trigger.

Two of the graded channels read a *difference between the two sides* of a bilaterally paired cell type, and there the standing asymmetry of a single reconstructed brain is a confound rather than a command: `DNa02` sits at 150 against 257 Hz and `DNp01` at 110 against 206 Hz, both of which exceed their command clamps and pin the button. We therefore subtract a slow baseline ($\tau = 3$ s) from the paired channels before decoding, keeping the transients and discarding the DC, and set each gain so that three standard deviations of what remains reach the clamp. The forward channel is not a bilateral pair — it is `BPN` − `MDN`, two opposing cell types — so its standing +27.6 Hz is a tonic walk command and is kept, with the gain placing it at half the clamp instead. We note the consequence rather than manage it: that difference carries ± 1.3 Hz of modulation on a 27.6 Hz mean, so walking speed here is 4.6% modulated and effectively tonic.

2.2 Gustatory pathway

As a positive control we reproduce the gustatory result of Shiu et al. [2024]. Stimulating sugar receptor neurons at 100 Hz drives the proboscis motor pool to 77 Hz; stimulating bitter receptors drives it to zero; stimulating both together yields 99% suppression relative to sugar alone.

This control has a known circularity that we state rather than manage: like Shiu et al. [2024] we fit W_{syn} on this very behaviour, obtaining 0.1655 mV against their 0.275 mV (the difference is attributable to a different connectome release and synapse pipeline). What the test genuinely establishes is therefore weaker than it appears — that *some* single scalar gain reproduces the behaviour — but it is not vacuous: the bitter–sugar suppression ratio was not fitted, and a sign-scrambled graph admits no such scalar.

2.3 Motion vision and looming

Direction selectivity fails. Presenting drifting gratings and measuring the direction selectivity index $\text{DSI} = (R_{\text{pref}} - R_{\text{null}})/(R_{\text{pref}} + R_{\text{null}})$ in `T4/T5`, the canonical elementary motion detectors [Maisak et al., 2013], gives $|\text{DSI}| \leq 0.0121$. Experimental studies routinely *select* terminals for analysis at $\text{DSI} > 0.5$, so the model sits some $40\times$ below the threshold at which a real terminal would merely qualify as direction-tuned.

Looming fails in the same way and more starkly (Table 2). Neither the intact nor a degree-preserving shuffled connectome distinguishes an expanding from a contracting stimulus; both instead show static $\gg$ looming $\approx$ receding, the signature of a response to dark *area* rather than to expansion.

Table 2. Looming (M4), open loop, mean rate in Hz. Neither arm separates looming from receding, and the intact connectome does not outperform a degree-preserving shuffle.

	Intact			**Shuffled**		
	`LC4`	`LPLC2`	`DNp01`	`LC4`	`LPLC2`	`DNp01`
Looming	36.06	0.00	139.0	7.04	2.82	88.7
Static	73.37	0.02	149.3	45.15	14.41	207.0
Receding	34.58	0.00	149.7	6.19	2.50	84.0

Because the shuffled graph performs comparably, the measurable looming response is not attributable to connectivity at all.

State of the looming detector

In closed loop `LPLC2` reads exactly 0.00 Hz, and the reason is measurable. It sits at -52.2 mV — *below* its own resting potential and 7.2 mV from threshold — because its heaviest input, `PVLP011` at -45.6 synapses per cell, fires at 371 Hz. The model's absolute refractory ceiling is $1/t_{\mathrm{refrac}} = 455$ Hz, so that one inhibitory cell runs at 82% of the maximum rate the model permits, and the inhibitory conductance it delivers exceeds the excitatory arm (`Tm5f`, $+44.7$ synapses per cell but only 36 Hz) by roughly tenfold. This is the `T4a` arm imbalance again, in a second and independent circuit.

That it is an imbalance rather than a shortage is what makes it diagnostic. We swept every input-side scale available — photoreceptor drive, the graded rate ceiling, and Weber contrast gain — and `LPLC2` does not leave 0.00 Hz at any setting. Reducing drive lowers `PVLP011` from 371 to 113 Hz but takes `LC4` from 53 to 0.0 Hz with it. The result is expected on inspection: an input scale multiplies both arms of a ratio and cancels, so no common gain can change a 10 : 1 imbalance. Only a gain that acts differentially on the two arms can, which is to say a per-type gain — the quantity the connectome does not specify and the optimized model of Lappalainen et al. [2024] obtains by fitting.

A tonic depolarising drive to the optic lobe does release it, and shows the same trade rather than escaping it. Raising the bias from 0 to 6 mV lifts `LPLC2` from 0.0 to 64.7 Hz, but it also lifts `MDN`, and the walking command is `BPN` $-$ `MDN`: that difference falls from $+22$ Hz to $+0.3$ Hz by 2 mV and reverses sign by 6 mV, at which point the model walks backwards. Meanwhile the population above 200 Hz grows from 197 cells to 991. There is no setting at which the looming pathway conducts and locomotion survives.

A blank-field control shows the trade is worse than that, and it is the reason we report bias sweeps with one. Presenting an *empty screen* at 7.5 mV bias drives `LPLC2` to 76.17 Hz, against 76.22 Hz for an expanding disc — a stimulus contribution of 0.07%. At 4 mV it is 30.52 blank against 30.62 looming. The bias that releases the cell is the same quantity that determines its output: past threshold we are no longer measuring vision but measuring the bias, and the fraction of the response attributable to the picture *falls* as the bias rises. There are two regimes and neither is looming detection — `LPLC2` silent, or `LPLC2` firing at a rate the stimulus barely perturbs. Reporting the looming response alone, without the blank, would have made 76 Hz look like success.

We also tested whether the failure is an artefact of where visual drive enters. By default we inject at the lamina monopolars, one synapse past the photoreceptors, which bypasses two real features of the eye: neural superposition, in which six photoreceptors sharing a visual direction converge on one cartridge, and the histaminergic first synapse that inverts the signal. Injecting at `R1-6` instead restores both. It changes nothing. At matched bias the two sites agree to within the blank-field offset (`LPLC2` 18.4 against 30.6 at 4 mV, 76.2 against 76.4 at 7.5 mV), and at zero bias photoreceptor injection is silent throughout, because the inhibition-dominated optic lobe has no baseline to disinhibit from. The one thing photoreceptor injection does carry is a bias of its own: FAFB's uneven optic-lobe proofreading recovers a column for 451 of 785 left columns against 749 of 796 right, so it is unusable for any measurement of a left–right difference, which is why the default sits at the lamina.

Across every condition in that sweep, looming and receding differ by less than the blank-field offset, and the *sign* of the difference flips between conditions. The only regime with genuine stimulus-driven

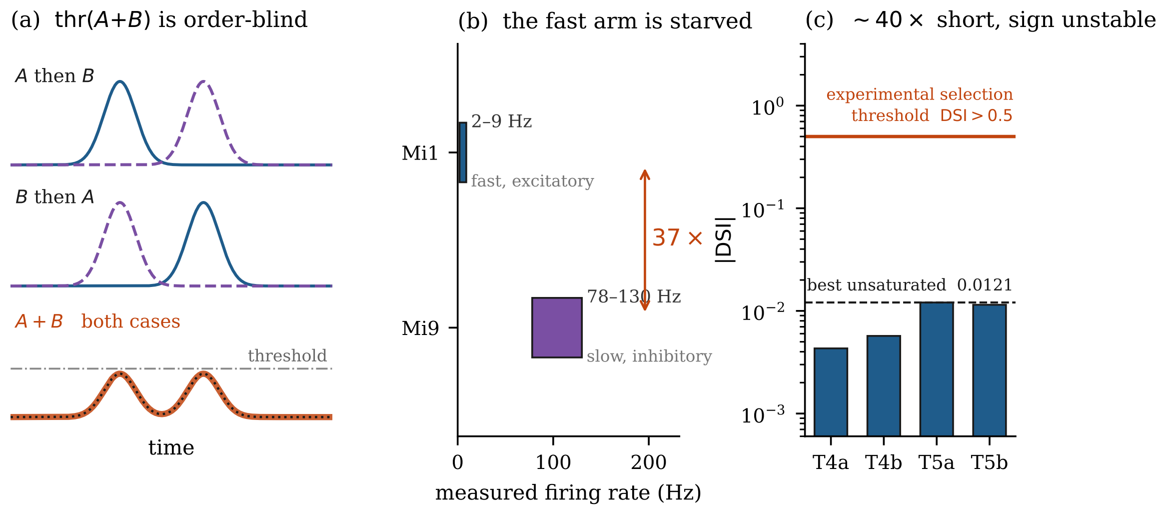

Figure 2. The motion failure. **(a)** The order-blindness argument we withdraw: for a *memoryless* unit, "A then B" and "B then A" produce identical sums (rust curve, second case overlaid dotted), so a threshold on that sum responds identically. This panel does not describe the present model, whose conduction delays and synaptic time constants make the summation history-dependent; a correlator built from the same units reaches DSI $= -0.79$ (Sec. 2.4). **(b)** Measured firing rates of `T4a`'s two correlator arms: the fast excitatory arm (`Mi1`) is suppressed by glutamatergic `L1` while the slow inhibitory arm (`Mi9`) runs freely, a $37\times$ conductance imbalance. **(c)** Resulting direction selectivity with graded optic-lobe units. Magnitudes fall some $40\times$ short of the value at which experimental studies merely *select* a terminal as direction-tuned, and across 45 unsaturated operating points the two mirror pairs oppose at only 3, which is below chance — the sign is fluctuating, not small-but-real.

structure is zero bias at the lamina, where `LC4` gives 77.2 Hz for a static disc against 24.9 Hz blank — a real threefold response, to dark *area*. The model reports how much dark is present, never whether it is growing.

Downstream visual functions fail correspondingly. `LPLC2` activity is uncorrelated with true angular size ($r = +0.000$), and no azimuth signal exists anywhere we could find: neither `LC11` ($r = +0.034$) nor `LC4` ($r = +0.070$) carries target bearing. `LC10a`, the cell type that would carry it, is silent at 0.00 Hz, receiving only $+45/-83$ synapses per cell against `LC11`'s $+477/-456$ and `LC4`'s $+439/-259$; it is held down by `AOTU042` and `TuTuAa`. Its known state-dependent gain boost is a *male* courtship mechanism gated by `P1`, and FAFB is a female brain, so that circuit is absent. Nor is there a state signal to substitute for it. `AOTU042` suppresses `LC10a` at -18 synapses per cell, but `pC1` and `AOTU042` are not connected in this brain at all — zero edges in either direction, at any synapse threshold — so the aggression state cannot reach that suppression to lift it or to deepen it.

2.4 Diagnosing the motion failure

Direction selectivity and looming selectivity are both computations over the *temporal order* in which neighbouring retinotopic columns activate. Motion to the right makes column A lead column B; motion to the left makes B lead A. The two cases contain identical events and differ only in order.

An earlier version of this manuscript argued that the failure was structural. In the model presynaptic inputs sum into a conductance and the only nonlinearity is a threshold on that sum; since $\mathrm{thr}(A + B)$ is symmetric in its arguments, we concluded that no unit of this class can distinguish the two cases. **That argument is wrong, and we withdraw it.** It holds only for a unit with no memory. The model has memory in two places the argument ignores: per-connection conduction delays, and a synaptic conductance that decays with $\tau_{\mathrm{syn}} = 5\,\mathrm{ms}$ rather than vanishing within the step. A delayed arm and a persistent conductance make the summand at any instant depend on stimulus history, and the symmetry does not hold.

We settled this by construction rather than by argument. Using the same neuron model, the same integrator and the same time step, we built a two-input correlator — two input units, one

Table 3. Direction selectivity, at the strongest operating point that is not saturated. Graded units are capped at 200 Hz and a unit against its ceiling reports DSI ≈ 0 whatever its inputs do, so we sweep bias, spatial period and temporal frequency (72 points), keep the 45 below 75% saturation, and report the largest effect among them — bias 1.0 mV, period 30°, 1 Hz. The magnitude is 41× below the value at which experimental studies merely *select* a terminal as direction-tuned. The sign is not reliable either: both mirror pairs oppose at only 3 of 45 unsaturated points (7%), by pair `T4a/T4b` 5/45, `T5a/T5b` 15/45. A correlator's sign should not depend on the operating point, so this is a quantity fluctuating about zero rather than a small but real selectivity.

Pathway	DSI		Mirror pair
ON	`T4a` -0.00431	`T4b` -0.00570	same sign
OFF	`T5a` -0.01208	`T5b` -0.01147	same sign

delayed, converging on a third — and measured it under the identical grating protocol. It reaches DSI $= -0.79 \pm 0.017$ (30 s $\times$ 5 seeds), against 0.002–0.016 for `T4a` in the network. The model class can compute direction selectivity to a magnitude far above the experimental selection threshold. Whatever prevents the network from doing so, it is not the arithmetic of summation and threshold. Biological elementary motion detectors combine a delayed and an undelayed arm *multiplicatively* [Hassenstein and Reichardt, 1956, Borst et al., 2010], realized in *Drosophila* as preferred-direction enhancement together with null-direction suppression [Haag et al., 2016, Gruntman et al., 2018]; a delay plus a saturating threshold approximates that product well enough to yield the 0.79 above.

Both ingredients are present in the graph, and we verified them. `T4a` receives fast centred excitation (`Mi1` $+30.6$, `Tm3` $+7.0$) and slow flanking inhibition on opposite sides (`Mi9` -7.4 at $(-0.76, +0.26)$; `Mi4` -3.4 at $(+0.39, -0.60)$), with four subtypes on four mirrored axes. The stimulus reaches them: `L1`'s response phase advances at -20.0 deg/deg against an expected 20.0, and this retinotopic travelling wave survives to `T4a`.

We implemented and measured six candidate remedies. Five produced nothing: per-type conduction delays (1.8–240 ms), per-neuron membrane time constants (5–200 ms), conductance-based shunting synapses, a forced high-conductance state ($g_{\text{tot}} = 105$, two orders of magnitude past the linear regime), and population pooling (per-cell DSI against per-cell input offset, $r = +0.043$ over 874 cells). The sixth — modelling the optic lobe as *graded, non-spiking* units, which is what photoreceptors and lamina monopolar cells actually are — was the only one to produce a measurable effect at all (Table 3), and it is small enough that its sign has to be checked rather than assumed.

Checking it requires sweeping the operating point rather than picking one, because the dominant influence on the number is not the wiring: graded units are capped at 200 Hz, and a unit against its ceiling reports DSI ≈ 0 whatever its inputs do. We therefore swept bias, spatial period and temporal frequency (72 points), kept the 45 below 75% saturation, and report the largest effect among them. Two things follow. The magnitude is $|\text{DSI}| \leq 0.0121$, some 40× below the value at which experimental studies merely *select* a terminal as direction-tuned. And the qualitative signature is absent: both mirror pairs oppose at only 3 of the 45 unsaturated points, where two independent coin flips would give about 11. A correlator's sign does not depend on its operating point, so what we are measuring is a quantity fluctuating about zero rather than a small but real selectivity. An earlier version of this manuscript reported opposed mirror pairs from a single configuration; the sweep does not support that claim and we withdraw it.

One imbalance is measurable and specific, though the tests below show it is not sufficient to explain the failure: the fast arm is starved. Weighting `T4a`'s inputs by measured firing rate rather than synapse count gives excitation 189 against inhibition 558, a 3.0× imbalance in drive; the realised synaptic conductances are $g_e = 0.0018$ against $g_i = 0.0667$, a ratio of 37×. The two quantities measure different things — rate-weighted input against the conductance it actually produces — and it is the second that sets how far the excitatory arm can move the membrane. `Mi1` sits at 2–9 Hz because `L1` suppresses it (-77.3, glutamatergic and therefore inhibitory), while `Mi9` runs at 78–130 Hz. The ON pathway is disinhibitory and, with uniform per-type gains, never rebalances. The OFF pathway, which does not depend on that disinhibition, conducts normally (`T5a` 58 Hz against `T4a` 2 Hz) — an internal control indicating the deficit is pathway-specific rather than a global failure of the optic lobe.

Correcting that imbalance does not rescue selectivity. We rebalanced the arms two ways: presynaptic short-term depression (Tsodyks–Markram, $U = 0.3$, $\tau_{rec} = 0.5\,s$), which depresses the tonically active inhibitory arm more than the sparsely active excitatory one; and transplanting the per-type gains of the optimized model of Lappalainen et al. [2024] onto the matching connections (571 of its 604 type-pairs align with our graph). Both *lowered* |DSI|. The 37× ratio is therefore a real property of the wiring under a single global gain, but it is not the operative cause, and the hypothesis we previously drew from it — that optimization supplies relative gain along the ON pathway's disinhibitory chain — is not supported by the transplant. We note the transplant is approximate: that model was fitted on a different reconstruction (fib25–fib19), so an unmatched gain is not evidence against the hypothesis, only against the simplest version of it.

Two further explanations were tested after the analysis above and are reported here because they are the two that a reader would reach for next.

The competing-input hypothesis. If the correlator is present but swamped, removing everything else should reveal it. We tested this in the RUNNING network rather than by isolated replay, which the caution at the end of this section explains cannot be trusted. Silencing every input onto `T4a/T4b` except `Mi1` and `Mi9` — 14,594 of 24,944 edges — does not raise selectivity: over the 45 operating points unsaturated in both conditions, max |DSI| moves from 0.0057 to 0.0020. Adding back `Tm3`, `Mi4` or `CT1` individually gives 0.0035, 0.0021 and 0.0034. `T5a/T5b` are never ablated and serve as a within-run control: their mean |DSI| is 0.00442–0.00444 in all five conditions, so nothing leaked. The manipulation is electrically real — `T4a`'s rate ranges 63 to 97 Hz across conditions. The correlator fails even when it is the only thing connected.

Dendritic compartmentalization. A point neuron makes every synapse electrically identical, so a shunt divides the whole cell, whereas null-direction suppression is local to a branch. We split all 11,822 `T4/T5` into a spiking soma and a passive dendrite coupled by an axial conductance, assigning each input by retinotopy alone — same column to the soma, different column to the dendrite. The rule reproduces the anatomy without naming a cell type (`Mi9` 97% distal, `Mi4` 98%) and is electrically effective (rates fall 193 → 151 Hz; unsaturated points rise 45 → 63). It does not rescue selectivity: max |DSI| 0.0121 → 0.0128, mirror-pair sign below chance throughout. Two compartments cannot represent the ordering of arms along a real `T4` dendrite, which is the obvious next refinement.

A related measurement bounds where the failure cannot be. At the tonic bias of 7.5 mV that makes an inhibition-dominated circuit conduct, the arms carry almost no time-varying signal at all: `Tm3` and `Mi4` sit at their ceiling 94.7% and 75.6% of the time with modulation amplitude 0.000 Hz, and `Mi9` delivers 0.93 Hz out of a 200 Hz range. A correlator has nothing to correlate there. The operating-point sweep already excludes that regime — its filter removes every point above 3 mV — but the filter tests `T4`'s own saturation, not its inputs', and those are different questions. Within the surviving range the two move together: as bias falls from 3.0 to 1.0 mV, arm modulation rises and max |DSI| rises with it, 0.0073 → 0.0121.

That trend does not extrapolate, and the way it fails is worth recording because it is a trap in the measure itself. Extending the sweep below the range previously tested, down to zero tonic drive, max |DSI| keeps rising, to 0.0164. It is not signal. `T4a` fires 2.0 Hz there, and the quantity a correlator actually delivers — the rate DIFFERENCE between preferred and null — falls over the same interval, from 0.31 Hz at 0.75 mV to 0.060 Hz at zero. The ratio grows because its denominator collapses faster than its numerator. The mirror-pair sign agrees: across these points the pairs share a sign on 78% of them, where a correlator requires opposition and chance would give 50%. DSI is therefore interpretable only in a middle band — bounded above by saturation, which the sweep already controls, and below by the vanishing denominator, which it did not. We report the bounded figure, $|\mathrm{DSI}| \leq 0.0121$, rather than the larger and emptier one.

What remains is a negative result without a mechanism, and we report it as such. Nineteen candidate explanations were measured and eliminated: pooling artefact (per-cell $r = +0.043$ over 874 cells); arm imbalance, by depression and by gain transplant; input dilution (the correlator supplies 80–86% of `T4a`'s input); unmodulated arms (`Mi1` modulates at 125% and `Mi9` at 89% of their means); `CT1` saturation (12,601 edges ablated, no change); absent phase offset (measured mean 154° between arms); smeared spatial sampling (inter-column coherence 1.000); same-column sampling (`Mi9/Mi4` sample one column away, 1.5% overlap); opposite-flank cancellation (flanks at 139°; ablating one does not help); graded versus spiking units; tonic bias 0–7.5 mV; global optic-lobe gain ×1–32; gain targeted onto `T4/T5` ×2–20; operating point, where the isolated correlator's monotonic DSI-versus-rate curve does not reproduce in the network; the competing-input hypothesis, by in-network ablation; and dendritic compartmentalization. The geometry, the delays, the phase offset and the modulation are all present and measured; the selectivity is not. We do not know why, and we prefer to say so.

Two properties of the model, rather than of the connectome, account for a large part of the deficit, and both were found by building a correlator that works and then adding `T4`'s properties to it one at a time.

The cell is below threshold on its own inputs. Summed over the graph, `T4a` receives 46.8 units of excitation and 41.5 of inhibition per cell. Calibrating the same units in isolation, a target driven at the graded ceiling emits nothing at a weight of 30 and 24 Hz at 60: the threshold for any output at all falls at roughly 40–50. Supplying the hand-built correlator with `T4a`'s measured weights therefore leaves it *silent*, $R_{\mathrm{pref}} = R_{\mathrm{null}} = 0$. Under a single global W_{syn} calibrated on the SEZ taste circuit, the cell can only be made to fire by the tonic bias, which carries no stimulus information — so the selectivity we have been measuring is a ripple on an uninformative pedestal. `T5a`, at 58.9 against 31.3, clears the same bar, and reads 3.7× the selectivity.

The graded transfer function removes the required nonlinearity. A correlator converts a phase difference between its arms into a rate difference, which requires an expansive nonlinearity. A graded unit emits $\mathrm{clamp}((v - V_{\mathrm{rest}})/\Delta, 0, 1)$, a *linear* ramp between its clamps. Holding the recorded `Mi1/Mi9` traces, the weights and the delays fixed and changing only the target's transfer function gives DSI $=$ -0.1781 spiking against -0.0168 graded, a 10.6× loss; the latter is the value the intact network reports. Modelling the optic lobe as graded is correct for photoreceptors and lamina monopolars and was adopted for that reason, but applied to `T4` it removes the operation the cell exists to perform.

Correcting both — spiking `T4/T5`, and optic-lobe gain in place of the tonic bias — raises the fraction of cells above the experimental selection threshold from 2.8% to 12.3%. It does not produce direction selectivity. The preferred direction remains set by the stimulus rather than by the wiring: across six spatial period and temporal frequency combinations, the fraction of `T5a` cells preferring one direction swings from 34% to 70%, crossing and reversing, and the residual response shows no delay tuning at all — varying the slow arm from 20 to 160 ms moves the directional signal only between 1.04 and 1.29 Hz. A correlator's preference is a property of its geometry and cannot depend on the stimulus that probes it. We therefore do not describe the residual as weak direction selectivity; it is a stimulus-timing artefact, which also explains the below-chance mirror opposition reported above.

One methodological caution follows from this work. An earlier draft reported mean $|\mathrm{DSI}| = 0.37$ for `T4a` cells driven by their own recorded inputs in isolation, and built a causal account on it. That measurement is an artefact and is withdrawn: the same procedure applied to a *single* input, where the geometry makes direction selectivity impossible, returns 0.12 rather than 0. The cause is that `Mi1` is itself weakly selective ($|\mathrm{DSI}| = 0.040$) and a threshold unit amplifies that residue. Isolated replay is not a clean control for this question.

2.5 Olfactory valence

The environment renders light and nothing else, but a fly in a room with a large animal in it smells that animal; simulating vision alone makes the model impoverished rather than conservative. We therefore added a channel feeding two real labelled lines — `ORN_DA1`, carrying the pheromone cVA [Kurtovic et al., 2007], and `ORN_DM1`, carrying vinegar [Semmelhack and Wang, 2009] — with enemies mapped to the former and health pickups to the latter.

The channel is deliberately built weak so that it cannot substitute for vision. Most importantly it carries *no direction*: a fly's antennae are ∼0.3 mm apart and cannot triangulate, so both sides receive bit-identical drive. It is additionally saturating, slow ($\tau = 0.2$ s), patchy (3 Hz bursts at 60% duty), and persistent ($\tau = 1.5$ s), so a source moving out of sight still smells.

Testing this in closed loop would be invalid (Section 2.6), so we ran it *open* loop: the agent is held still and the game stepped with a null action, so enemies still move and the scene still evolves, but it evolves identically in both arms because the seed is fixed and nothing feeds back. Smell is then the

Table 4. Effect of the olfactory channel, open loop, 10 seeds × 320 tics. Δ is the mean on-minus-off difference in Hz; "sign" counts seeds agreeing with the mean direction, with p from a two-sided sign test. In the intact connectome every motor and valence population moves consistently; under a degree-preserving shuffle of the same graph every effect falls to the 0.00–0.11 Hz scale of the `LC4` control. Sign consistency alone does not separate the arms — the shuffle retains some of it — so the comparison to read is magnitude. `LC4` is a visual cell whose activity should be unaffected by smell; it carries a small offset in both arms, at 10479× smaller magnitude than the lateral horn response it bounds.

	Intact				**Degree-preserving shuffle**			
Population	Δ (Hz)	s.d.	sign	p	Δ (Hz)	s.d.	sign	Role
`LHN`	+14.06	5.46	10/10	0.002	+0.01	0.01	8/10	innate valence
`DNp01`	−17.67	7.42	9/10	0.021	−0.00	0.27	6/10	escape (giant fibre)
`DNa02`	+11.68	5.29	9/10	0.021	+0.11	0.34	7/10	steering
`BPN`	+1.21	0.46	10/10	0.002	+0.07	0.04	10/10	forward walking
`MDN`	−0.96	0.40	10/10	0.002	−0.02	0.15	6/10	backward walking
`LC4`	−0.00	0.02	8/10	0.109	+0.07	0.03	10/10	*visual control*

only difference between arms, and `LC4` — a visual cell that should not move — serves as a built-in check that the comparison is sound.

The lateral horn — the innate-valence pathway [Jefferis et al., 2007, Dolan et al., 2019] — is dormant at 0.32 Hz without smell and rises to 14.1 ± 5.5 Hz with it, in all ten seeds (Table 4). The effect reaches the descending neurons in *one synapse*, and its sign is predicted by the wiring: `LH` → `DNp01` is −248 synapses, hence inhibitory, and `DNp01` is suppressed by 17.7 ± 7.4 Hz, in nine seeds of ten. Meanwhile `LC4`, a visual population that should not move at all, does not: it shifts by -0.00 ± 0.02 Hz, four orders of magnitude below the lateral horn response it bounds.

Crucially, a degree-preserving shuffle of the same graph, run through the identical protocol and the identical seeds, does not reproduce this (Table 4, right). The valence and steering effects collapse by two to three orders of magnitude — `LHN` from +14.06 to +0.01 Hz (1563×), `DNa02` 102×, `DNp01` to within noise of zero — landing at the 0.00–0.11 Hz scale of the `LC4` control, that is, at the network's generic coupling floor.

Sign consistency and effect magnitude discriminate differently here, and only the second separates the arms. Sign consistency does *not*: several shuffled populations remain fairly sign-consistent across seeds, including `LC4` at 10/10 — the one population that should not be responding at all. What discriminates is magnitude. A sign test on a 0.01 Hz effect is a statement about a systematic that is three orders of magnitude too small to matter, and and we interpret it accordingly. The shuffled `DNp01` residual reversed sign between two independent executions of the protocol, where every intact effect held its direction — the behaviour expected of noise rather than of a transformation. In summary, the intact effects are large and reproducible while the shuffled ones sit at the control's floor. *The effect requires the wiring.*

The `BPN` and `MDN` walking effects support a weaker claim than the rest. They are consistent in direction across all ten seeds, but at ±1 Hz they are only ∼17× above the shuffled floor rather than ∼100×, and we draw no conclusion from them in isolation.

The shuffle also changes the baseline in an informative way. Without smell the intact lateral horn sits at 0.32 Hz; the shuffled one idles at 25.7–25.8 Hz across seeds, an 80× higher floor. Preserving degree while destroying specificity does not merely remove the response — it removes the *quiet*. The intact lateral horn is selectively silent and selectively driven, which is the behaviour a labelled-line valence pathway should show, and neither property survives rewiring.

Two caveats apply to this result throughout. Odour source distances are read from the engine's label buffer, so the model was *told* that enemies exist; this is evidence about what the connectome does with a signal, not evidence that it can detect one. And 1,227 lateral horn neurons pooled together mixes many functional channels, so "the lateral horn responds" is coarser than it sounds.

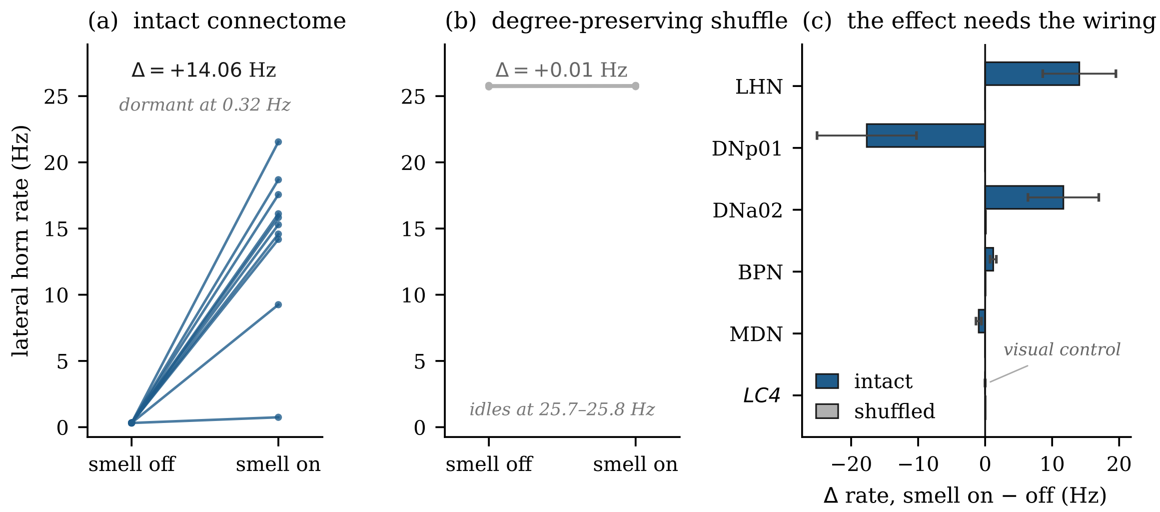

Figure 3. Olfactory valence reaches the descending neurons, and needs the wiring to do it. **(a)** Per-seed lateral horn rate with smell off and on in the intact connectome; all ten seeds rise. **(b)** The same protocol on a degree-preserving shuffle of the same graph: no seed moves appreciably, and the shuffled baseline idles near 21 Hz where the intact pathway is dormant at 0.32 Hz. **(c)** Mean change per population, both arms (error bars are s.d. over ten seeds). `LC4` is a visual cell included as a control: it should not move, and does not.

Synaptic weight against path length

We first routed the same odour at the `pC1` aggression neurons, on the strength of a three-hop path touching 8 of 10 cells, and it failed flat — 0.00 Hz across a tenfold sweep of drive. Measured afterwards, the reason is structural: `DA1`'s contribution is a rounding error against `pC1`'s balanced $+530/-555$ of input, and `pC1` is a bistable latch that must be pushed over rather than nudged. The lateral horn is the opposite on every count — it receives 21% of the odour relay's entire output, has no latch, and sits one hop from the output. Hop count is a poor proxy for influence in a connectome; synaptic weight against competing input is the quantity that matters.

2.6 Validity of shuffle controls in closed loop

Degree-preserving shuffles are the standard control for attributing an effect to connectivity, and we use one above. But in a closed loop they silently stop being a control.

We measured this directly. Over one matched episode the intact agent encountered an enemy on 186 of 500 tics and finished at full health; the shuffled agent encountered one on 489 of 492 tics and finished at 16 health. The point this episode makes is qualitative — that the arms visit different states — and it does not require the difference to be of any particular size; but see the power note below before reading episode counts as effects. The two arms behaved differently, and because behaviour determines what the agent sees, they were no longer sampling the same stimulus distribution. Any cross-arm correlation then compares two different worlds rather than two different brains.

The consequence generalizes to any embodied connectome study: *controls must hold the stimulus fixed*, which in practice means running the comparison open loop, as we do for the olfaction result.

3 Discussion

The dissociation we observe is not between easy and hard computations, nor between small and large circuits. It is between computations that a linear summation followed by a threshold can express and those it cannot. Gustatory and innate-olfactory valence are, at the level the connectome constrains them, weighted sums with signs: sugar excites a motor pool, bitter inhibits it, an odour drives a valence region that inhibits an escape neuron. Those transfer. Motion direction is a product of a delayed and an undelayed signal, and it does not.

This result is easily over-read as a claim about connectomics, and we state its scope precisely. The connectome contains the correlator: we found the geometry, the mirrored axes, the phase offset and the travelling wave that drives them. What the frozen model does not supply is whatever additional structure makes those ingredients function together, and we are careful not to name it. Our first candidate was the per-type gain that makes the two arms comparable in magnitude — in our hands they differ by 37× — but correcting the imbalance, by depression and by transplanting an optimized model's gains, lowered selectivity rather than raising it. The honest summary is that a simulator demonstrably capable of DSI = 0.79 in a hand-built correlator yields 0.002 from the same units wired by the connectome, and we cannot presently say what the difference consists of.

Placed beside Lappalainen et al. [2024], this becomes informative rather than merely negative. That model shares our wiring and recovers motion selectivity, but obtains its unknown parameters by optimization. Ours fixes them to a single published set. The difference between the two outcomes is therefore a measurement of what the optimization supplies. We took the natural next step and inspected it: of that model's 604 per-type-pair gains, 571 align with connections in our graph, and transplanting them lowers our selectivity rather than raising it. Gain alone, at least as transferred between two different reconstructions, is not what is missing. What that leaves — timing structure, dendritic compartmentalization absent from a point neuron, or an interaction between gains that does not survive transplantation — we cannot separate with the present model.

Two further readings should be excluded. First, none of this is a claim about whether a fly brain can play a video game; the environment is outside the animal's evolutionary experience and the model has no ventral nerve cord, so behavioural competence was never the measurable. Second, the olfaction result is not evidence of enemy detection — the odour magnitude is supplied by the engine. What both results measure is transfer: whether a transformation implied by the wiring appears in a running system.

3.1 Limitations

- **No ventral nerve cord.** FAFB is brain-only, so the game's movement code substitutes for legs. All behavioural *timescales* are the engine's; claims are restricted to descending-neuron activity.
- **No gap junctions.** EM reconstruction captures chemical synapses; the giant fibre's fastest output is electrical and therefore invisible here.
- **No neuromodulation.** Dopamine, octopamine and serotonin are flattened to fast excitation.
- $n = 1$ **brain, and its halves are not mirror images.** `DNa02` sits $\sim 2\times$ higher on the right than the left, which is a standing turn command until removed by a slow baseline.
- **Angular scale is calibrated, not measured.** The lattice is real but its axes are not isotropic in visual angle, so each is scaled to the published field of view. Absolute angular claims inherit this.
- **The gustatory control is partly true by construction**, since W_{syn} is fitted on it.
- **Single stimulus environment.** All visual results come from one rendering engine with one viewport geometry; we have not tested whether a different visual statistic changes the arm imbalance.

4 Methods

4.1 Connectome data

FlyWire FAFB v783 [Dorkenwald et al., 2024, Schlegel et al., 2024], frozen October 2023. Connectivity from the Buhmann synapse pipeline [Buhmann et al., 2021] with transmitter predictions from Eckstein et al. [2024]. Edges thresholded at five synapses. Cell types from the consolidated typing table; `BPN` and `aIPg` exist only in the free-text community label file. The data are CC BY-NC 4.0 and are not redistributed with our code.

4.2 Neuron model

Leaky integrate-and-fire with parameters as listed in Section 2.1. The subthreshold system $\tau \dot{v} = (V_{\text{rest}} - v) + g$, with g decaying at τ_{syn}, is integrated exactly:

$$v(t+dt) = v_0 A + g_0 B + g_{\text{ext}}(1 - A), \qquad A = e^{-dt/\tau_{\text{mem}}}, \quad B = \frac{\tau_{\text{syn}}}{\tau_{\text{syn}} - \tau_{\text{mem}}}\left(e^{-dt/\tau_{\text{syn}}} - A\right),$$

matching the scheme used by Brian 2 [Stimberg et al., 2019], against which we validate to 0.03%.

4.3 Departures from the reference model

Each is switchable and each carries a measured justification. (i) *Histamine override* on photoreceptor output, as above. (ii) *Conductance-based synapses*: $\tau\dot{v} = (V_{\text{rest}} - v) + g_e(E_e - v) + g_i(E_i - v) + I$ with $E_e = 0$, $E_i = -70\,\text{mV}$, verified divisive (shunting reduces input–output slope by $> 40\%$ where subtraction preserves it). (iii) *Per-type conduction delays* via a multi-tap ring buffer, since a single global delay cannot express a correlator. (iv) *Graded non-spiking units* for all 77,873 optic neurons, output rectified-linear in $(v - V_{\text{rest}})$ capped at 200 Hz. (v) *Weber contrast adaptation* per column. (vi) `L3` kept sustained while `L1/L2` adapt — from a stationary agent the rendered scene changes by 0.0003 in mean luminance between frames, so a fully adapting retina washes out and the loop cannot start.

4.4 Visual input and the environment

Modifying the game is not a modification of the model, and three properties of Doom are wrong for this animal in ways that suppress the signal under test. *Gamma*: the engine writes gamma-encoded sRGB, but photoreceptors integrate linear flux and the acceptance function is a spatial integral over it, so the Gaussian pre-blur must run in linear light; undoing the encoding first raises retinal contrast by 37%. *Spectrum*: `R1-6` carries Rh1, which is blue-green sensitive and nearly blind at Doom's red primary, while Doom is painted in browns — weighting the frame by R1-6 sensitivity therefore *lowers* retinal signal relative to human luma. *Palette*: because `R1-6` are a single spectral class the motion pathway is monochromatic, so hue is not a quantity it can use, and we re-emit the scene's achromatic structure in blue-green — equivalent to repainting every surface, and the same move a fly-vision laboratory makes when it builds an arena from green emitters. That recovers the lost signal and then some, raising retinal standard deviation from 0.020 to 0.033, measured open loop on 100 identical frames.

None of it releases `LPLC2`. We note the engine's own display controls are not a route either: the engine's gamma, contrast, brightness and saturation controls all leave the rendered buffer bit-identical, because ViZDoom exposes it upstream of display post-processing, and `r_visibility`, which does take effect, moves frame mean brightness without changing the distance-to-brightness relation by more than 3–6% over a 40-tic approach. Every one of these is a common scale on both arms of the imbalance, which is why the ratio argument predicts the outcome in each case.

4.5 Rate readout and motor decoding

An exponential filter over per-step output. Because that output is rate $\times\, dt$, recovering Hz requires division by dt; dividing by τ instead scales every reported rate by 160$\times$ and is a failure mode we hit and document. A 12-tic warmup precedes baseline estimation, and a $\tau = 3\,\text{s}$ baseline removes the standing left–right asymmetry.

4.6 Statistical analysis

The olfaction comparison uses 10 independent environment seeds. Each seed contributes one paired on/off difference, averaged over 320 tics after a 40-tic settling period that lets the rate filter converge. We report the mean and standard deviation across seeds together with a two-sided sign test on the number agreeing with the mean direction ($p = 2\sum_{i=k}^{n}\binom{n}{i}/2^n$; $p = 0.002$ at 10/10, $p = 0.021$ at 9/10). We prefer the sign test to a t-test because per-seed effect magnitude depends on how closely enemies happen to approach, which is neither normal nor identically distributed across seeds — it is a property of the episode, not of the model. Note that sign consistency alone is a weak claim when an effect is small, which is why we read the `LC4` control on magnitude rather than on its p-value. The degree-preserving shuffle permutes postsynaptic targets while preserving each neuron's in-degree, and is run through the identical protocol with the identical seeds. Table and figure are generated from the same serialized run data, so they cannot diverge.

The environment is not bit-reproducible across processes, so the magnitudes carry run-to-run variation of a few percent and the sign counts are less stable than a single execution suggests. We can put a number on the latter: correcting an error in the odour-source table — `CustomMedikit`, the pickup actor used by one scenario, was absent from the food list, and unlisted actors fell through to the pheromone channel, so inanimate scenery was being smelled as conspecifics — left the lateral horn response essentially unchanged in size but moved `DNp01` and `DNa02` from 10/10 agreeing seeds to 9/10. Sign counts near ceiling should therefore be read as p values with one seed of slack, not as invariants.

Power of episode-level behavioural measures. Behaviour in this environment is far noisier than the open-loop rate measurements, and we flag the consequence because it constrains what any closed-loop connectome study can claim. Health collected per episode has a standard deviation of roughly 13 units across environment seeds, so at $n = 5$ only differences of about 20 units are resolvable; two executions of one identical configuration returned means of 3.2 and 11.2. Detecting a 5-unit difference at 80% power would require $n \approx 106$. We therefore report no behavioural effect from fewer than 30 seeds, quote 95% confidence intervals rather than standard deviations for these measures, and treat the interval rather than the point estimate as the result. Rate-level measurements, which are paired and open loop, are not subject to this and are reported as above.

4.7 Tested and deferred

Recording what was tried and failed is as much a part of the result as what worked, so we list both.

Tested, and does not rescue visual motion: per-type conduction delays (1.8–240 ms); per-neuron membrane time constants (5–200 ms); conductance-based shunting synapses, which are now the default rather than a candidate; a forced high-conductance state; population pooling; graded non-spiking optic-lobe units; tonic optic-lobe bias from 0 to 7.5 mV; photoreceptor drive, graded rate ceiling and Weber contrast gain; injection at the photoreceptors rather than the lamina; field of view from 110° to 170°; sRGB linearisation; R1-6 spectral weighting; repainting the environment into the fly's band; and the engine's own rendering controls. Most of these share a structure — each is a scale applied to both arms of a ratio, and a scale cancels — but that account is incomplete, because the *asymmetric* interventions listed below, which do not cancel, failed as well.

Corrected during this work, all on the input side: the retina mapped azimuth linearly onto a perspective projection, misplacing a column at 10° of gaze by 87%; the field of view was never applied, because ViZDoom ignores it as a command-line argument, so the model ran at the engine default rather than the reported value; frames were held across all 57 substeps of a tic, presenting a correlator with an impulse train aliased against its own 80 ms delay line; and, consequently, Weber adaptation advanced once per tic rather than once per substep, giving it an effective time constant of 14 s in place of 0.25 s. Each of these corrupts precisely the cue under test. None of them changed the outcome, which is itself informative: the negative result does not rest on a badly delivered stimulus.

Tested in this round, and also negative: presynaptic short-term depression as an arm rebalancer; transplantation of 571 per-type gains from an optimized model; ablation of `CT1` (12,601 edges); ablation of one inhibitory flank; global optic-lobe gain to $\times 32$ and gain targeted onto `T4`/`T5` to $\times 20$; and matching the network's firing-rate regime to that of the isolated correlator. *Positive control added:* a hand-built two-input correlator in the same simulator reaches DSI $= -0.79 \pm 0.017$, which is what licenses us to call the network result a failure of the wired model rather than of the method. Rebuilt from scratch for this revision, the same construction reaches -0.26 at one configuration and -1.00 once the real fan-in and spatial spread are added; we quote the original figure but note that the magnitude depends on the operating point chosen, and that the qualitative claim — this model class computes direction — is the part the argument rests on.

A defect in our own instrumentation, disclosed because it invalidates earlier statements. The `--spiking-t4` option was applied in only one of three code paths; the per-cell and operating-grid paths build their own graded mask and silently ignored it. Any conclusion about spiking `T4` drawn from those paths measured the graded model twice, and is withdrawn. The option is now applied in one place used by all three. Separately, an earlier per-subtype geometry analysis pooled `Mi9` with `Mi4`, which sit on *opposite* flanks; averaging them produced a vector that flips with their relative weight and manufactured both a spurious bimodality and an apparent 97–114° mirror-pair defect. Taking `Mi9` alone, the axes are coherent ($R = 0.67$–0.93) and the mirror pairs are 182–194° apart, as they should be. `T4`'s input geometry is not defective.

Deferred. Per-cell-type synaptic gains remain unfitted, because fitting them is the independent variable of this study rather than a repair to it — a position unchanged by the transplant result above, which tests transfer rather than fitting. We note the target is not a single set of numbers: the optimized model of Lappalainen et al. [2024] carries 734 free parameters, 604 of them one gain per type-to-type connection, and its authors report an ensemble of 50 solutions of comparable task performance, so connectome plus task does not identify them uniquely. Also deferred: gap junctions, absent from the reconstruction and carrying the giant fibre's fastest output; neuromodulation, flattened here to fast transmission; and a ventral nerve cord, without which every behavioural timescale is the engine's.

Data availability

The connectome data are FlyWire FAFB v783 [Dorkenwald et al., 2024, Schlegel et al., 2024], licensed CC BY-NC 4.0. They are not redistributed with this work and must be obtained from FlyWire directly; a fetch script is provided with the code. The per-cell-type gains used in the transplant ablation are derived from the published `flyvis` ensemble of Lappalainen et al. [2024]. All serialized run data behind the tables and figures are included in the code repository, and every table and figure is generated from those files rather than transcribed, so the two cannot diverge.

Code availability

Code is available at `https://github.com/mutkuoz/flydoom` under the MIT licence, including the simulator, all ten experiments, and the scripts that generate every table and figure. The environment uses ViZDoom [Kempka et al., 2016] with the freely licensed Freedoom asset set, so no commercial game data are required.

DOOM is a trademark of id Software LLC / ZeniMax Media Inc. This work is unaffiliated with and unendorsed by them, and the name is used here only to identify the software environment.

Author contributions

M.U.Ö. conceived the study, wrote the simulation and analysis code, ran the experiments, and wrote the manuscript.

Competing interests

The author declares no competing interests.

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